ECO6102: Problem Set 1
(Due on Sunday 11 October 2026)

Instructions

Submission. Please use Python to answer the programming questions in this problem set. Submit **two files** on https://bb.cuhk.edu.cn/: (1) a Jupyter notebook (`.ipynb`) containing your Python code, and (2) a PDF file containing your answers to both the theoretical and the programming questions. Be sure to include your name and student ID in both files. Points may be deducted if these instructions are not followed.

Deadline. Problem sets are due by 23:59 on the date shown. **No late assignments will be accepted**.

Data. Question 4 uses `attend.csv`, posted on Blackboard with this problem set. Report the usual OLS standard errors (the `statsmodels` default) unless a question asks otherwise. If you use the Stata file (`attend.dta`) instead, load it with `pd.read_stata(...).astype(float)`: some variables are stored as one-byte integers, and squaring them overflows without any warning.

Sources. All questions are from Wooldridge (2010), *Econometric Analysis of Cross Section and Panel Data*, 2nd edition, restated here in the course notation. The bracket beside each question gives the chapter, problem number and page; part 4(g) is not from the book.

Use of AI tools. You may use AI tools (for example ChatGPT, DeepSeek, Kimi or Copilot), and we encourage you to **use them critically**: AI is often wrong in econometrics, so check its algebra and code, write your answers in your own words, and make sure you can explain every step. You are responsible for what you submit. **Question 5, a short AI-use statement, is required in every submission.** It is **not graded**, and honest use costs no points.

Conceptual Questions

Question 1 [Wooldridge (2010), Ch. 2, Problem 2.2(a)–(b), p. 28]

Let y and x be scalar random variables, and let $\mu = E(x)$. Suppose the conditional mean of y is a quadratic in x around its mean:

$$E(y \mid x) = \delta_0 + \delta_1(x - \mu) + \delta_2(x - \mu)^2.$$

(a) Derive $\partial E(y \mid x)/\partial x$. How does it depend on x?

(b) Show that δ_1 equals $\partial E(y \mid x)/\partial x$ averaged over the distribution of x.

Question 2 [Wooldridge (2010), Ch. 4, Problem 4.7, p. 84]

You want to estimate how owning a personal computer affects college grades. Let $PC = 1$ if a student owns a personal computer and $PC = 0$ otherwise, and let *colGPA* be college GPA. You also observe high school GPA and SAT scores, and specify

$$colGPA = \beta_0 + \beta_1 hsGPA + \beta_2 SAT + \beta_3 PC + u.$$

(a) Give reasons why u and PC might be positively correlated.

(b) Suppose the equation is estimated by OLS from a random sample of college students. Is $\hat{\beta}_3$ more likely to have an upward or a downward asymptotic bias? Explain.

(c) What variables might serve as good proxies for the unobservables in u that are correlated with PC?

Question 3 [Wooldridge (2010), Ch. 4, Problem 4.8(a)–(c), pp. 84–85]

Consider a population regression with two explanatory variables that interact, with x_2 also entering as a quadratic:

$$E(y \mid x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \beta_4 x_2^2.$$

Let $\mu_1 \equiv E(x_1)$ and $\mu_2 \equiv E(x_2)$ be the population means.

(a) Let α_1 be the average partial effect of x_1 on $E(y \mid x_1, x_2)$, averaged over the distribution of the explanatory variables, and let α_2 be the same for x_2. Find α_1 and α_2 in terms of the β_j and μ_j.

(b) Rewrite the regression function so that α_1 and α_2 appear directly as coefficients. (μ_1 and μ_2 will also appear.)

(c) Given a random sample, what regression would you run to estimate α_1 and α_2 directly? What if you do not know μ_1 and μ_2?

Programming Questions

Question 4 [Wooldridge (2010), Ch. 4, Problem 4.14, p. 86; part (g) added]

Data: `attend.csv`, $N = 680$ students in one course. Variables used:

stndfnl	standardized final exam score
atndrte	percentage of lectures attended
frosh, *soph*	$= 1$ for a first-year, second-year student
priGPA	cumulative GPA before the term
ACT	ACT achievement test score

(a) To estimate the effect of attending lectures on final exam performance, regress *stndfnl* on *atndrte*, with *frosh* and *soph* as additional explanatory variables. Interpret the coefficient on *atndrte* and discuss its statistical significance.

(b) How confident are you that the OLS estimates in part (a) measure the causal effect of attendance? Explain.

(c) As proxies for student ability, add *priGPA* and *ACT* to the regression. What is the effect of *atndrte* now? Discuss how it differs from part (a).

(d) What happens to the significance of the dummy variables in part (c), compared with part (a)? Explain.

(e) Add $priGPA^2$ and ACT^2 to the regression. What happens to the coefficient on *atndrte*? Are the quadratics jointly significant?

(f) To test for a nonlinear effect of *atndrte*, add $atndrte^2$ to the regression in part (e). What do you conclude?

(g) Let m be the sample mean of *atndrte*. Re-estimate the regression in (f), replacing *atndrte* and $atndrte^2$ by $(atndrte - m)$ and $(atndrte - m)^2$. Compare the coefficient on the linear attendance term and its standard error with (f). What does each coefficient measure? Relate your answer to Question 1.

AI-Use Statement

Question 5 [required in every submission; not graded]

Required. Answer briefly, 3–5 lines in total, at the end of your PDF. If you used no AI tools, write "No AI tools used" and stop there.

(a) Which AI tools and models did you use?

(b) Which questions did you use them for, and how?

(c) What did the AI get wrong that you caught and corrected?

(d) What kind of task did it struggle most to answer accurately?

Example. *"(a) ChatGPT (GPT-5). (b) To explain a Python error in Question 4 and to check my algebra in Question 3(b). (c) Its first answer to 3(b) dropped a term, which I fixed. (d) It was reliable on Python syntax but struggled most with interpreting regression output."*